



## Handout 6.7: Quick Bread Lab Sheet

Name: \_\_\_\_\_ Date: \_\_\_\_\_ Period: \_\_\_\_\_

Station: \_\_\_\_\_ My role: \_\_\_\_\_

Print for every student. Bring it both days. Day 1 is pages 1 to 3, Day 2 is pages 4 and 5.

## DAY 1

### Part 1. How leavening works

Write it in four sentences, in your own words.







**Which one needs an acid to work? Circle: baking powder / baking soda**

**Where does the acid come from in a real recipe? Name three sources:**

1. \_\_\_\_\_ 2. \_\_\_\_\_ 3. \_\_\_\_\_

**Part 2. The four-cup fizz test**

| Cup | What is in it           | What happened (none, a little, a lot) |
|-----|-------------------------|---------------------------------------|
| 1   | Baking soda + water     |                                       |
| 2   | Baking soda + vinegar   |                                       |
| 3   | Baking powder + water   |                                       |
| 4   | Baking powder + vinegar |                                       |

**The rule, in one sentence:**

Baking soda needs \_\_\_\_\_, and baking powder does not because

## Part 3. The experiment setup

### The four batches

| Batch       | Leavening treatment        | What I predict |
|-------------|----------------------------|----------------|
| A (control) | 1 tsp baking powder        |                |
| B           | none at all                |                |
| C           | 2 tsp baking powder        |                |
| D           | 1 tsp baking soda, no acid |                |

### Variables

**Independent variable** (the one thing we change on purpose):

**Dependent variable** (what we measure to see the result). Name all three:

1. \_\_\_\_\_ 2. \_\_\_\_\_ 3. \_\_\_\_\_

**Constants** (everything we keep exactly the same). List at least four, and go for eight:

|    |    |
|----|----|
| 1. | 5. |
| 2. | 6. |
| 3. | 7. |
| 4. | 8. |

**Why do we need Batch A at all, if we already know a normal muffin rises?**





#### Part 4. My hypothesis and prediction

**My hypothesis** (one sentence, in the form “if we change , *then* ”):





**My prediction: rank the four batches by how high I think they will rise.**

Highest \_\_\_\_ second \_\_\_\_ third \_\_\_\_ lowest \_\_\_\_

**Batch D prediction:** what will it taste like and why?



## Day 1 exit card

The one thing we are changing is \_\_\_\_\_.

Batch \_\_\_\_\_ will rise the highest because \_\_\_\_\_.

## DAY 2

### Part 5. Data

Measure the batter height in the liner before baking and the muffin height after. Use the ruler.

| Batch                | Height<br>BEFORE<br>baking | Height<br>AFTER<br>baking | How much it<br>rose | Crumb<br>(none, few,<br>or many<br>holes) | Taste |
|----------------------|----------------------------|---------------------------|---------------------|-------------------------------------------|-------|
| A (control)          |                            |                           |                     |                                           |       |
| B (no<br>leavening)  |                            |                           |                     |                                           |       |
| C (double)           |                            |                           |                     |                                           |       |
| D (soda, no<br>acid) |                            |                           |                     |                                           |       |

**Time the tin went in:** \_\_\_\_ **Time it came out:** \_\_\_\_ **Oven temperature:** \_\_\_\_

#### Actual ranking by height

Highest \_\_\_\_ second \_\_\_\_ third \_\_\_\_ lowest \_\_\_\_

#### Compare to your prediction

| Batch | I predicted | What actually<br>happened | Same or different? |
|-------|-------------|---------------------------|--------------------|
| A     |             |                           |                    |
| B     |             |                           |                    |
| C     |             |                           |                    |
| D     |             |                           |                    |

### Part 6. Conclusion

Fill in all four parts.

**The data showed** \_\_\_\_\_



**My hypothesis was** (circle one) **supported** / **not supported** / **partly supported**, because



**My evidence is** \_\_\_\_\_





**One thing that could have gone wrong with our test is \_\_\_\_\_**



**One sentence our station can read out loud at the station check:**



### Station check

| Line                                                          | Done? |
|---------------------------------------------------------------|-------|
| Oven off, tin on the trivet, nothing hot left out             |       |
| All four bowls washed, dried, and put away per the drawer map |       |
| Measuring cups and spoons washed and put away                 |       |
| Counters wiped and sanitized                                  |       |
| Sink empty and rinsed                                         |       |
| Floor swept, trash out                                        |       |
| Leavening jars returned to the teacher's table                |       |

Teacher signature: \_\_\_\_\_

## Grade 8 stretch: Batch E

**Batch E: 1 tsp baking soda PLUS 1 tsp white vinegar.** Everything else the same.

| Batch E | Before | After | How much it rose | Crumb | Taste |
|---------|--------|-------|------------------|-------|-------|
|         |        |       |                  |       |       |

**Does Batch E behave more like Batch A or more like Batch D? \_\_\_\_\_**

**What does that prove? \_\_\_\_\_**



**Name one confounding variable in our own test** (something we did not control that could have affected the result), **and say how you would control it next time.**

The variable: \_\_\_\_\_

How to control it: \_\_\_\_\_

## Teacher key

**Part 1, how leavening works:** (1) Baking powder and baking soda both make carbon dioxide gas when they get wet and hot. (2) The gas makes bubbles inside the batter. (3) The bubbles get trapped and push the batter up. (4) The heat of the oven sets the batter around the bubbles, so the bread stays risen after it cools.

**Baking soda** needs the acid. Acid sources in real recipes: buttermilk, yogurt, sour cream, vinegar, lemon juice, brown sugar, molasses, honey, cocoa (natural, not Dutch processed), applesauce, cream of tartar. Any three.

### Part 2, fizz test results:

| Cup                 | Result   | Why                                                                                          |
|---------------------|----------|----------------------------------------------------------------------------------------------|
| 1. Soda + water     | a little | Water alone triggers almost nothing. A faint fizz at most                                    |
| 2. Soda + vinegar   | a lot    | Vinegar is the acid. This is the overflow cup                                                |
| 3. Powder + water   | a lot    | Baking powder already contains a dry acid, so water is enough                                |
| 4. Powder + vinegar | a lot    | It reacts, but not dramatically more than with water, because it did not need the extra acid |

**The rule:** baking soda needs an acid, and baking powder does not, because baking powder already has its own acid mixed in with it.

### Part 3, expected predictions and actual results:

| Batch             | Expected result                                                                                          | What usually happens                                                                           |
|-------------------|----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| A (control)       | Normal rise, even crumb, tastes right                                                                    | The benchmark. About 1 1/2 to 2 inches tall with a domed top                                   |
| B (none)          | Barely rises, dense, heavy, gummy                                                                        | Comes out flat and pale, about half the height of A. Edible but tough. Tastes fine, just dense |
| C (double)        | Rises fastest and highest, then often sinks in the middle; large uneven holes; may taste slightly bitter | The best data in the experiment. Too much gas builds a structure that cannot hold itself up    |
| D (soda, no acid) | Rises a little, and tastes soapy, metallic, or bitter;                                                   | The unreacted baking soda is what students taste. Tell                                         |

|  |                                          |                                                                      |
|--|------------------------------------------|----------------------------------------------------------------------|
|  | may look slightly yellow or green inside | them before they taste so nobody thinks something is wrong with them |
|--|------------------------------------------|----------------------------------------------------------------------|

**Constants students should list:** flour, sugar, salt, milk, oil, egg, how much batter goes in each liner, the bowl, the amount of stirring, the oven, the oven temperature, the bake time, the shelf in the oven, the liners, the room.

**Why Batch A exists:** without a normal batch baked in the same oven on the same day, there is nothing to compare the other three to. The control is the ruler.

**Part 6, a strong conclusion:** “The data showed that leavening is what makes a quick bread rise, and that more is not better. Batch A rose about 1 3/4 inches, Batch B rose about 3/4 of an inch, Batch C rose about 2 inches and then sank in the middle, and Batch D rose about 1 inch and tasted soapy. My hypothesis was partly supported, because I predicted C would be the tallest and it was, but I did not predict it would collapse. My evidence is the height column and the crumb column. One thing that could have gone wrong is that our Batch B bowl got stirred longer than the others, which makes any batter tougher.”

**Grade 8, Batch E:** it behaves like Batch A. It rises well, because the vinegar gave the baking soda the acid it needed. That proves the missing ingredient in Batch D was the acid and not the soda, which is the clean end of the experiment.

**Grade 8, confounding variables worth credit:** different students stirred different amounts; the four liners did not get exactly the same amount of batter; one side of the oven runs hotter; the liners were not all in the same row; the egg was not divided exactly evenly; one batch sat longer before going in the oven, which matters because baking powder starts working as soon as it gets wet.

**Scoring:** this lab sheet carries all three lesson objectives and is scored with Rubric 01 - Lab Rubric.md, three objective rows plus the safety and sanitation row, 16 points. Objective 3 is scored from Part 6 of this sheet. Objectives 1 and 2 and safety are scored on the clipboard during Day 2.